# Analysis I

## Bsc. Informatik ETH Zürich

Laurin Seeholzer

May 12, 2026

# Contents

# 1 Logic, Sets and Numbers

## 1.1 Logic

**Definition 1.1.** *A **mathematical Statement** is a well-formulated assertion that is strictly either true or false.*

**Example 1.2.** *"Every prime number is odd." is a false statement.*

**Definition 1.3.** *A **Statement** that depends on one or multiple variables is called a **predicate** and denoted by $P(x), Q(x), \ldots$.*

**Example 1.4.**

$$P(x) := x > 2$$

**Definition 1.5.** *The negation of a mathematical Statement $A$ is defined as $\neg A :=$ $A$ is false.*

**Example 1.6.** $A := \forall x \in \mathbb{R} : x > 2$ $\neg A := \exists x \in \mathbb{R} : x \leq 2$

**Definition 1.7.** ***Quantifiers*** *are short hand notation for making statements about predicates.*

$$\forall x \in \mathbb{R} : P(x) \iff \text{for all} x \in \mathbb{R} \text{ holds } P(x)$$

$$\exists x \in \mathbb{R} : P(x) \iff \text{there exists} x \in \mathbb{R} \text{ s.t. } P(x)$$

$$\exists! x \in \mathbb{R} : P(x) \iff \text{there exists exactly one} x \in \mathbb{R} \text{ s.t. } P(x)$$

$$\nexists x \in \mathbb{R} : P(x) \iff \text{there exists no} x \in \mathbb{R} \text{ s.t. } P(x)$$

**Example 1.8.**

$$\forall x \in \mathbb{R} : x^2 \geq 0$$

$$\forall x \in \mathbb{R} \exists y \in \mathbb{R} : x = y^2$$

**Important 1.9.** *The oderder of quantifiers matters. $\forall x \exists y$ is not equivalent to $\exists y \forall x$.*

**Definition 1.10.** *Basic **logical connectives** are defined as:*

$$A \wedge B := A \text{ and } B \text{ are true}$$

$$A \vee B := A \text{ or } B \text{ is true}$$

$$A \implies B := A \text{ implies } B \text{ is true}$$

$$A \iff B := A \text{ is equivalent to } B \text{ is true}$$

**Remark 1.11.** *Observe that equivalence holds iff.* $A \implies B \wedge B \implies A$

**Lemma 1.12.** *From the above definition it follows that:*

$$\neg A \vee B \iff A \implies B$$

*This can be shown by evaluating the truth table for both statements.*

**Theorem 1.13.** *The following equivalences hold:*

$$\neg(\forall x P(x)) \iff \exists x \neg P(x)$$

$$\neg(\exists x P(x)) \iff \forall x \neg P(x)$$

*aswell as*

$$\forall x(P(x) \wedge Q(x)) \iff (\forall x P(x)) \wedge (\forall x Q(x))$$
$$\exists x(P(x) \vee Q(x)) \iff (\exists x P(x)) \vee (\exists x Q(x))$$

## 1.2 Sets

**Definition 1.14.** *A* ***set*** *is a collection of elements, that are listed or characterized by a certain property. The empty set is denoted by* $\emptyset$ *and is the only set that contains no elements.*

**Definition 1.15.** *An element* $x$ *is* ***contained in a set*** $A$ *iff.* $x \in A$ *else it is not contained in* $A$ *denoted by* $x \notin A$.

**Definition 1.16.** *Let* $A$ *and* $B$ *be sets.* $A$ *is a* ***subset*** *of* $B$ *denoted by* $A \subset B$ *iff.*

$$\forall x(x \in A \implies x \in B)$$

$A$ *is a* ***proper subset*** *of* $B$ *denoted by* $A \subsetneq B$ *iff.*

$$A \subset B \wedge A \neq B$$

**Definition 1.17.** *Let $A$ and $B$ be sets.*

- *The **union** of $A$ and $B$ is defined as $A \cup B \coloneqq \{x \mid x \in A \vee x \in B\}$.*
- *The **intersection** of $A$ and $B$ is defined as $A \cap B \coloneqq \{x \mid x \in A \wedge x \in B\}$.*
- *The **difference** of $A$ and $B$ is defined as $A \setminus B \coloneqq \{x \mid x \in A \wedge x \notin B\}$.*
- *The **Cartesian product** of $A$ and $B$ is defined as $A \times B \coloneqq \{(a, b) \mid a \in A \wedge b \in B\}$.*

**Remark 1.18.** *Note that if $B \subset A$ then the diffrence $A \setminus B$ is called the complement of $B$ in $A$ and is denoted by $B^c$.*

### 1.3 Numbers

The basic and often used sets of numbers are:

- $\mathbb{N}$: The set of natural numbers 1, 2, 3, 4, ...
- $\mathbb{Z}$: The set of integers ... -2, -1, 0, 1, 2, ...
- $\mathbb{Q}$: The set of rational numbers $\frac{p}{q}$ where $p, q \in \mathbb{Z}$
- $\mathbb{R}$: The set of real numbers

**Remark 1.19.** *Some important subsets of the real numbers are $\mathbb{R}_0^+/\mathbb{R}^+$ and $\mathbb{R}_0^-/\mathbb{R}^-$ which define the intervalls $[0, \infty)/(0, \infty)$ and $(-\infty, 0]/(-\infty, 0)$.*

**Definition 1.20.** *Continuous subsets are called **intervalls**. An intervall is called*

- ***open** if it does not contain its endpoints $I = (a, b) \coloneqq \{x \mid a < x < b\}$*
- ***closed** if it contains its endpoints $I = [a, b] \coloneqq \{x \mid a \leq x \leq b\}$*
- ***half-open** if it contains one endpoint and not the other $I = [a, b) \coloneqq \{x \mid a \leq x < b\}$*

**Remark 1.21.** *An arbitrary subset of $\mathbb{R}$ is called open if it is a union of open intervalls and closed if its complement is open.*

**Definition 1.22.** *An **upper (lower) bound** of a subset $X \subset \mathbb{R}$ is an element $y \in \mathbb{R}$ such that:*

$$\forall x \in X : x \leq (\geq) y$$

*The **maximum (minimum)** of a subset $X \subset \mathbb{R}$ is the element $x_0 \in X$ such that:*

$$\forall x \in X : x \leq (\geq) x_0$$

*Often denoted as **max(X)** and **min(X)**.*

**Definition 1.23.** *An intervall $I = (a, b)$ is **bounded** if there exist upper and lower bounds.*

**Remark 1.24.** *A bounded and closed intervall is often refered to as **compact**.*

**Definition 1.25.** *The **Supremum** $sup(X)$ of a subset $X \subset \mathbb{R}$ is the least upper bound of $X$. and can be characterized by:*

$$\forall x \in X : x \leq sup(X) \wedge \forall \epsilon > 0 : \exists x \in X : x > sup(X) - \epsilon$$

**Definition 1.26.** *The **Infimum** $inf(X)$ of a subset $X \subset \mathbb{R}$ is the greatest lower bound of $X$. and can be characterized by:*

$$\forall x \in X : x \geq inf(X) \wedge \forall \epsilon > 0 : \exists x \in X : x < inf(X) + \epsilon$$

**Example 1.27.** *Let $X = [-2, 1)$ we have a minimum of $x_0 = -1$ but no maximum. Observe that it is bounded from above by $1$ and $2$ or any number larger than $1$ but only $1$ is the least upper bound (Supremum).*

**Theorem 1.28.** *The **maximum and minimum are unique** aslong as they exist.*

*Proof.* Assume there exists $x_1, x_2 \in X$ such that $x_1 = max(X)$ and $x_2 = max(X)$. As both $x_1$ and $x_2$ are maximum of $X$ we have

$$x_1 \geq x_2 \wedge x_2 \geq x_1 \implies x_1 = x_2$$

□

**Definition 1.29.** ***Axioms of Addition***

- *Commutativity: $x + y = y + x$*
- *Associativity: $x + (y + z) = (x + y) + z$*
- *Identity element: There exists an element $e \in X$ such that $x + e = x$ for all $x \in X$.*
- *Inverse element: For every $x \in X$ there exists an element $x^{-1} \in X$ such that*

$x + x^{-1} = e$.

**Remark 1.30.** *A set $X$ together with an operation $+$ that satisfies the axioms of addition is called a* ***abelian group****.*

**Example 1.31.** $(\mathbb{Z}, +)$ *is an abelian group.*

**Definition 1.32.** ***Axioms of Multiplication***

- *Commutativity:* $x \cdot y = y \cdot x$
- *Associativity:* $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- *Identity element: There exists an element $e \in X$ such that $x \cdot e = x$ for all $x \in X$.*
- *Inverse element: For every $x \in X$ there exists an element $x^{-1} \in X$ such that $x \cdot x^{-1} = e$.*

**Definition 1.33.** ***Distributivity*** *of multiplication over addition:*

$$\forall x, y, z \in X : x \cdot (y + z) = x \cdot y + x \cdot z$$

**Remark 1.34.** *A set $X$ together with operations $+$ and $\cdot$ that satisfies the axioms of addition and multiplication and distributivity is called a* ***field****.*

**Definition 1.35.** ***Order Axioms***

- *reflexivity:* $\forall x \in X : x \leq x$
- *antisymmetry:* $\forall x, y \in X : x \leq y \wedge y \leq x \implies x = y$
- *transitivity:* $\forall x, y, z \in X : x \leq y \wedge y \leq z \implies x \leq z$
- *totality:* $\forall x, y \in X : x \leq y \vee y \leq x$

**Definition 1.36.** *A set $X$ is said to be* ***(order)complete*** *iff for every non empty subsets $A, B \subseteq X$*

$$\forall a \in A \forall b \in B : a \leq b$$
$$\implies \exists c \in X \quad s.t. \quad \forall a \in A : a \leq c \wedge \forall b \in B : c \leq b$$

**Example 1.37.** *The real numbers $\mathbb{R}$ satisfy axiomatic properties for addition, multiplication, distributivity and order. Furthermore $\mathbb{R}$ is an order complete field.*

**Example 1.38.** *$\mathbb{Q}$ is not order complete. Let $A = \{x \in \mathbb{Q} \mid x^2 \leq 2\}$ and $B = \{x \in \mathbb{Q} \mid x^2 \geq 2\}$. Then $\forall a \in A \forall b \in B : a \leq b$ but there is no $c \in \mathbb{Q}$ such that $c^2 = 2$.*

**Theorem 1.39.** *Every **non empty, bounded from above (below)** subset $X \subset \mathbb{R}$ has a **supremum (infimum)**.*

*Proof.* Let $B$ be the set of all upper bounds of $X$. As $X$ is bounded from above and non empty, $B$ is non empty. Therfore

$$\forall x \in X : x \leq b \quad \forall b \in B$$

By the order completenes of $\mathbb{R}$ there exists a $c \in \mathbb{R}$ such that

$$\forall x \in X : x \leq c \wedge \forall b \in B : c \leq b$$

Hence $c$ is the least upper bound of $X$ and therefore $c = sup(X)$. Assume there exists $c' \in \mathbb{R}$ such that $c' = sup(X)$ and $c' \neq c$. Then by definition of $sup(X)$ we have

$$c \leq c' \wedge c' \leq c \implies c = c'$$

□

**Theorem 1.40.** *The **Archimedian Principle** says that*

- *For $x \in \mathbb{R}$ and $y > 0$ there exists an element $n \in \mathbb{N}$ such that $n * y > x$.*
- *For every $\epsilon > 0$ there exists an element $n \in \mathbb{N}$ such that $\frac{1}{n} < \epsilon$.*

*Proof.* We prove the first theorem by contradition. Assume it does not hold hence there exists $x \in \mathbb{R}$ and $y > 0$ such that

$$\forall n \in \mathbb{N} : n * y \leq x$$

This implies that $x$ is an upper bound for the set $X = \{n * y \mid n \in \mathbb{N}\}$. As $X$ is non empty and bounded from above, it has a supremum $c = sup(X)$. As $y > 0$ $c - y$ can not be an upper bound of $X$ hence there exists $n_0 \in \mathbb{N}$ such that

$$c - y < n_0 * y \implies c < (n_0 + 1) * y$$

This contradicts the assumption that $c$ is the supremum of $X$. □

*Proof.* The second theorem is a direct consequence of the first theorem. It can be shown by setting $x = 1$ in the first theorem. □

**Theorem 1.41.** *The Young Inequality states that for all $\epsilon > 0$ and all $x, y \in \mathbb{R}$:*

$$2|xy| \leq \epsilon x^2 + \frac{1}{\epsilon} y^2$$

*Proof.*

$$\left(\sqrt{\epsilon} x - \frac{1}{\sqrt{\epsilon}} y\right)^2 \geq 0 \implies \epsilon x^2 - 2xy + \frac{1}{\epsilon} y^2 \geq 0 \implies \epsilon x^2 + \frac{1}{\epsilon} y^2 \geq 2xy$$

□

### 1.3.1 Vectors and Vector Spaces

**Definition 1.42.** *The* ***euclidian space*** $\mathbb{R}^n$ *is the set of all vectors* $x = (x_1, x_2, \ldots, x_n)$ *where* $x_i \in \mathbb{R}$.

**Definition 1.43.** ***Addition*** *and* ***scalar multiplication*** *are defined as:*

$$x + y = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n)$$

$$\lambda x = (\lambda x_1, \lambda x_2, \ldots, \lambda x_n)$$

**Definition 1.44.** *A* ***vector space*** *has to satisfy the following axioms:*

- *Commutativity:* $x + y = y + x$
- *Associativity:* $x + (y + z) = (x + y) + z$
- *Identity element: There exists an element* $e \in X$ *such that* $x + e = x$ *for all* $x \in X$.
- *Inverse element: For every* $x \in X$ *there exists an element* $x^{-1} \in X$ *such that* $x + x^{-1} = e$.
- *Distributivity:* $x \cdot (y + z) = x \cdot y + x \cdot z$

**Definition 1.45.** *For* $x, y \in \mathbb{R}^n$*, the* ***standard scalar product*** *is defined as:*

$$x \cdot y = \langle x, y \rangle := \sum_{i=1}^{n} x_i y_i$$

*The* ***euclidean norm*** *is defined as:*

$$||x|| := \sqrt{\sum_{i=1}^{n} x_i^2}$$

*The* ***euclidean distance*** *between* $x$ *and* $y$ *is:*

$$d(x, y) := \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

**Theorem 1.46.** *The triangle- and the cauch-schwarz inequality hold for all* $x, y, z \in$

$\mathbb{R}^n$:

$$||x - z|| \leq ||x - y|| + ||y - z||$$
$$|\langle x, y\rangle| \leq ||x|| ||y||$$

### 1.3.2 Complex Numbers

**Definition 1.47.** *The* ***imaginary unit*** $i$ *is defined as* $i^2 = -1$.

**Definition 1.48.** *The* ***complex numbers*** $\mathbb{C}$ *are the set of numbers*

$$\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$$

*where* $a$ *is the* ***real part*** $Re(z)$, *and* $b$ *is the* ***imaginary part*** $Im(z)$.

**Definition 1.49.** *Arithmetic of complex numbers* $z_1 = x_1 + iy_1$ *and* $z_2 = x_2 + iy_2$:

- *Addition:* $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$
- *Multiplication:* $z_1 \cdot z_2 = (x_1x_2 - y_1y_2) + i(x_1y_2 + y_1x_2)$

**Definition 1.50.** *The* ***complex conjugate*** *of* $z = x + iy$ *is* $\bar{z} = x - iy$.

**Important 1.51.** *Division is calculated by expanding with the conjugate of the denominator:*

$$\frac{z}{w} = \frac{z \cdot \bar{w}}{w \cdot \bar{w}} = \frac{z \cdot \bar{w}}{|w|^2}$$

**Remark 1.52.** *Some usfull properties of the conjugate:*

- $z \cdot \bar{z} = |z|^2$
- $z + \bar{z} = 2Re(z)$ *and* $z - \bar{z} = 2iIm(z)$
- $\overline{z + w} = \bar{z} + \bar{w}$ *and* $\overline{z \cdot w} = \bar{z} \cdot \bar{w}$

**Definition 1.53.** *Geometrically, complex numbers can be represented in the Gaussian plane. The* ***absolute value*** *of* $z = a + ib$ *is its* ***distance from the origin***:

$$|z| = \sqrt{a^2 + b^2}$$

*The distance between $z_1$ and $z_2$ is $d = |z_1 - z_2|$.*

**Remark 1.54.** *Observe that the triangle inequality holds in $\mathbb{C}$: $|z + w| \leq |z| + |w|$.*

**Theorem 1.55.** *The Euler's formula states that for $t \in \mathbb{R}$:*

$$e^{it} = \cos(t) + i\sin(t)$$

*This allows writing trigonometric functions using complex exponentials:*

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}, \quad \sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

**Remark 1.56.** *The **Intuition** behind eulers formula is we can use Taylor series to show that:*

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

*Using $x = it$ we get:*

$$e^{it} = \sum_{n=0}^{\infty} \frac{(it)^n}{n!} = \cdots = \cos(t) + i\sin(t)$$

Eulers formula leads us to a new representation of complex numbers: **the polarcoordinates**.

**Definition 1.57.** *For $z \in \mathbb{C}$, the polar-coordinates are defined as:*

$$z = r(\cos(t) + i\sin(t)) = re^{it}$$

*where $r = |z|$ is the absolute value and $t = \arg(z)$ is the argument.*

**Remark 1.58.** *To convert from cartesian to polar coordinates:*

$$r = \sqrt{a^2 + b^2}, \quad t = \arctan\left(\frac{b}{a}\right)$$

*Note that the arctan function only returns values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so we have to adjust the angle based on the quadrant of the complex number.*

**Example 1.59.** *Multiplication and division in polar-coordinates is much easier than in cartesian coordinates:*

$$z_1 \cdot z_2 = r_1 e^{it_1} \cdot r_2 e^{it_2} = r_1 r_2 e^{i(t_1 + t_2)}$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{it_1}}{r_2 e^{it_2}} = \frac{r_1}{r_2} e^{i(t_1 - t_2)}$$

**Example 1.60.** *So is exponentiation or taking roots:*

$$z^n = (re^{it})^n = r^n e^{int}$$

$$\sqrt[n]{z} = \sqrt[n]{r} e^{i\frac{t+2\pi k}{n}}, \quad k \in \{0, 1, \ldots, n-1\}$$

**Definition 1.61.** *The $n$-th roots of unity are the solutions to $z^n = 1$, which are given by:*

$$z_k = e^{i\frac{2\pi k}{n}}, \quad k \in \{0, 1, \ldots, n-1\}$$

**Theorem 1.62.** *The sum of the roots of unit is always zero.*

$$\sum_{k=0}^{n-1} e^{i\frac{2\pi k}{n}} = 0$$

**Theorem 1.63.** *The* ***fundamental theorem of algebra*** *states that every polynomial of degree $n \geq 1$ has exactly $n$ roots in $\mathbb{C}$ (counting multiplicity). These roots appear in konplex-conjugate pairs.*

# 2 Sequences and Series

## 2.1 Sequences

**Definition 2.1.** *A **sequence** (Folge) $(a_n)_{n\in\mathbb{N}_0}$ is an infinite list of numbers, mathematically a mapping $f : \mathbb{N}_0 \to X$. It can be given explicitly or recursively as*

$$a_n = f(n) \quad \textit{or} \quad a_n = g(a_{n-1}, a_{n-2}, \dots)$$

**Example 2.2.** *A famous example is the Fibonacci sequence defined by $a_0 = 0, a_1 = 1$ and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$. The first few terms are $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$*

**Definition 2.3.** *A sequence $(a_n)$ is called **convergent** with limit $L \in \mathbb{R}$ if:*

$$\forall \epsilon > 0 \exists N > 0 \quad s.t. \quad \forall n > N : |a_n - L| < \epsilon$$

*Written as $\lim_{n\to\infty} a_n = L$.If no such $L$ exists, the sequence is **divergent**.*

**Remark 2.4.** *A special case of divergent sequences are sequences which diverge to $\pm\infty$. Their limits are called **improper limits**.*

**Theorem 2.5.** *The limit of a sequence is unique aslong as it exists.*

*Proof.* Assume $(a_n)$ converges to $L_1$ and $L_2$. Then for any $\epsilon > 0$ there exists $N_1$ and $N_2$ such that:

$$|a_n - L_1| < \epsilon \quad \forall n > N_1$$
$$|a_n - L_2| < \epsilon \quad \forall n > N_2$$

Then for any $n > \max(N_1, N_2)$:

$$|L_1 - L_2| = |L_1 - a_n + a_n - L_2| \leq |L_1 - a_n| + |a_n - L_2| < 2\epsilon$$

Since this holds for any $\epsilon > 0$, we must have $L_1 = L_2$. □

**Theorem 2.6.** *Any subsequence $(a_{n_k})_{k\in\mathbb{N}_0}$ of a convergent sequence converges to the same limit $L$.*

**Definition 2.7.** *An **accumulation point** (Häufungspunkt) $A \in \mathbb{R}$ of a sequence fulfills:*

$$\forall \epsilon > 0 \forall N \in \mathbb{N}_0 \exists n \geq N \textit{ such that } |a_n - A| < \epsilon$$

**Example 2.8.** *The sequence $a_n = \frac{1}{2}(-1)^n$ has the accumulation points $A_1 = \frac{1}{2}$ and $A_2 = -\frac{1}{2}$.*

**Theorem 2.9.** *A is an accumulation point iff there exists a convergent subsequence with limit A.*

*Proof.* Let $(a_n)$ be a sequence and $A$ an accumulation point. We can easily construct a subsequence $(a_{n_k})$ that converges to $A$ by selecting $n_k$ such that $|a_{n_k} - A| < 1/k$ this is given to exist by the definition of an accumulation point. Now let $(a_{n_k})$ be a convergent subsequence with limit $A$ then for any $\epsilon > 0$ there exists $K$ such that $|a_{n_k} - A| < \epsilon$ for all $k > K$ and since $n_k \to \infty$ as $k \to \infty$ by definition it follows that $A$ is an accumulation point. □

**Theorem 2.10.** *Every convergent sequence has exactly one accumulation point (its limit).*

*Proof.* For $A$ to be an accumulation point of $a_n$ we need that there are indefinitely many $a_n$ that approach $A$. If $a_n$ converges to $L$ then we can intuitively say that that $A$ has to be $L$. □

**Definition 2.11.** *A sequence is* ***upper (lower) bounded*** *if there exists M such that* $|a_n| \leq (\geq) M \quad \forall n \in \mathbb{N}_0$.

**Theorem 2.12.** *Every convergent sequence is bounded.*

*Proof.* By the definition of convergence there exists $N$ sucht that for all $n \geq N$ we have $|a_n - L| < 1$. Thus $|a_n| < |L| + 1$ for all $n \geq N$. Further we have a finite number of terms $a_0, \dots, a_{N-1}$ which are also bounded. Therefore the sequence is bounded by $M = \max(|a_0|, \dots, |a_{N-1}|, |L| + 1)$. □

**Theorem 2.13.** ***Bolzano-Weierstrass Theorem:*** *Every bounded sequence has an accumulation point.*

**Theorem 2.14.** *Limit arithmetic: For* $\lim a_n = K$, $\lim b_n = L$ *and* $C \in \mathbb{R}$:

- $\lim(a_n \pm b_n) = K \pm L$
- $\lim(C \cdot a_n \cdot b_n) = C \cdot K \cdot L$
- $\lim(a_n / b_n) = K/L$ *(if* $L \neq 0$ *and* $b_n \neq 0$*)*

**Remark 2.15.** *Important limits:*

- $\lim_{n\to\infty} \frac{\ln n}{n} = 0$

- $\lim_{n\to\infty} n^{1/n} = 1$
- $\lim_{n\to\infty} x^{1/n} = 1 \quad (x > 0)$
- $\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$
- $\lim_{n\to\infty} \frac{x^n}{n!} = 0$

**Theorem 2.16.** ***Sandwich-Theorem:*** *If $a_n \leq b_n \leq c_n$ for $n \geq N_0$, and $\lim a_n = \lim c_n = L$, then $\lim b_n = L$.*

*Proof.* Let $\lim a_n = \lim c_n = L$ and $\lim b_n = M$. From the above arithmetic we can sa that $a_n \leq b_n \leq c_n$ implies $L \leq M \leq L$ so we have that $b_n$ is squeezed between $a_n$ and $c_n$ and thus must converge to $M = L$. $\square$

**Example 2.17.** *We want to calculat $\lim_{n\to\infty}(2^n + 5^n)^{\frac{1}{n}}$. We can bound the sequence by $5^n \leq 2^n + 5^n \leq 2 \cdot 5^n$. Taking the $n$-th root we get $5 \leq (2^n + 5^n)^{\frac{1}{n}} \leq 2^{\frac{1}{n}} \cdot 5$. As $n \to \infty$ we have $2^{\frac{1}{n}} \to 1$ so by the sandwich theorem we have $\lim_{n\to\infty}(2^n + 5^n)^{\frac{1}{n}} = 5$.*

**Remark 2.18.** *The sandwich theorem is often used to find upper and lower bounds for a sequence that converge to the same limit and therfore conclude that the original sequence does too.*

**Definition 2.19.** *A sequence is* ***monotonically increasing (decreasing)*** *if $a_n \leq (\geq) a_m$ for $m > n$.*

**Theorem 2.20.** ***Monotone Convergence Theorem:*** *Every bounded monotonic sequence converges to its supremum (infimum).*

*Proof.* Wlog let $(a_n)$ be a bounded monotonically increasing sequence. Hence there exists $S = \sup\{a_n : n \in \mathbb{N}_0\}$ such that $a_n \leq S$ for all $n$ and for any $\epsilon > 0$ there exists $N$ such that $a_N > S - \epsilon$ and therfore $|a_n - S| < \epsilon$ for all $n \geq N$. $\square$

**Definition 2.21.** *The limit of the supremum is called* ***limes superior*** *and denoted as:*

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup_{k\geq n} a_k$$

*The limit of the infimum is called* ***limes inferior*** *and denoted as:*

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf_{k\geq n} a_k$$

**Definition 2.22.** *A sequence is a* ***Cauchy sequence*** *iff*

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : |a_n - a_m| < \epsilon$$

**Theorem 2.23.** *Every Cauchy sequence is bounded.*

*Proof.* Choose $\epsilon = 1$, then there exists $N$ such that $|a_n - a_N| < 1$ for all $n \geq N$. Thus $|a_n| < |a_N| + 1$ for $n \geq N$, meaning $(a_n)$ is bounded by $M = \max(|a_0|, \ldots, |a_{N-1}|, |a_N| + 1)$. □

**Theorem 2.24.** *A sequence of real numbers converges iff it is a cauchy sequence.*

*Proof.* $\Longrightarrow$ : Let $(a_n)$ converge to $L$. For any $\epsilon > 0$, there exists $N$ such that $\forall n \geq N$ : $|a_n - L| < \epsilon/2$. Then for any $m, n \geq N$:

$$|a_n - a_m| = |a_n - L + L - a_m| \leq |a_n - L| + |L - a_m| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus, $(a_n)$ is a Cauchy sequence.

$\Longleftarrow$ : Let $(a_n)$ be a Cauchy sequence therfore it is bounded. By the Bolzano-Weierstrass theorem, the bounded sequence $(a_n)$ has an accumulation point $L$, so there exists a subsequence $(a_{n_k})$ converging to $L$. For any $\epsilon > 0$, since $(a_n)$ is Cauchy, there exists $N'$ such that $\forall m, n \geq N' : |a_n - a_m| < \epsilon/2$. Since $a_{n_k} \to L$, choose a $k$ such that $n_k \geq N'$ and $|a_{n_k} - L| < \epsilon/2$. Then for any $n \geq N'$:

$$|a_n - L| \leq |a_n - a_{n_k}| + |a_{n_k} - L| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

Thus, the sequence converges to $L$. □

## 2.2 Series

**Definition 2.25.** *A* ***series*** *(Reihe) is an infinite sum $\sum_{i=0}^{\infty} a_i$ of the terms of a sequence $(a_n)$.*

**Definition 2.26.** *A series is said to converge iff the sequence of partialsums converges.*

$$\sum_{i=0}^{\infty} a_i = \lim_{n \to \infty} \sum_{i=0}^{n} a_i$$

**Remark 2.27.** *Some examples of series are:*

- ***geometric series:*** $\sum_{n=0}^{\infty} x^n$ *converges to* $\frac{1}{1-x}$ *iff* $|x| < 1$
- ***harmonic series:*** $\sum_{n=1}^{\infty} \frac{1}{n}$ *converges for* $s > 1$

**Theorem 2.28.** ***Necessary condition for convergence:*** *for* $\sum_{i=0}^{\infty} a_i$ *to converge we need* $\lim_{n\to\infty} a_n = 0$ *i.e. it is a null sequence.*

**Theorem 2.29.** *For* ***series with non-negative terms*** $a_n \geq 0$*, the sequence of partial sums is monotonically increasing and* ***converges iff it is bounded.***

**Definition 2.30.** *A series is* ***absolutely convergent*** *if* $\sum |a_n|$ *converges. A* ***conditionally convergent*** *series converges, but not absolutely.*

**Remark 2.31.** *Absolute convergence implies normal convergence.*

**Theorem 2.32.** ***Riemann's rearrangement theorem:*** *Let* $\sum a_n$ *be a conditionally convergent series. Then for any* $x \in \mathbb{R}$*, there exists a rearrangement of the series that converges to* $x$*.*

**Theorem 2.33.** ***Comparison test*** *(Vergleichskriterium): If* $c_n \leq b_n \leq a_n$ *for* $n \geq N$*:*

- *If* $\sum a_n$ *converges, then* $\sum b_n$ *converges (**Majorantenkriterium**).*
- *If* $\sum c_n$ *diverges, then* $\sum b_n$ *diverges (**Minorantenkriterium**).*

**Example 2.34.**

$$\sum_{n=1}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \ldots$$

*We can compare this series to the geometric series*

$$1 + \sum_{n=1}^{\infty} (\frac{1}{2})^n = 1 + 2 = 3$$

*Therfore accordint to the Majorantenkriterium the series* $\sum_{n=1}^{\infty} \frac{1}{n!}$ *converges.*

**Theorem 2.35.** ***Leibniz Criterion:*** *For an alternating series $\sum(-1)^n a_n$ with a monotonically decreasing null sequence $(a_n)$, the series converges.*

*Proof.* Let $(a_n)$ be a non-negative, monotonically decreasing sequence converging to 0. Let $s_n = \sum_{k=0}^{n}(-1)^k a_k$ be the $n$-th partial sum. For the even partial sums $(s_{2n})$, we have

$$s_{2n+2} - s_{2n} = -a_{2n+1} + a_{2n+2} \leq 0$$

since $a_{2n+2} \leq a_{2n+1}$. Thus, $(s_{2n})$ is monotonically decreasing. Similarly for the odd partial sums $(s_{2n+1})$, we have

$$s_{2n+3} - s_{2n+1} = a_{2n+2} - a_{2n+3} \geq 0$$

meaning $(s_{2n+1})$ is monotonically increasing. Note that

$$s_{2n+1} = s_{2n} - a_{2n+1} \leq s_{2n} \leq s_0 = a_0$$

This limits both bounds: $0 \leq s_{2n+1} \leq s_{2n} \leq a_0$. By the Monotone Convergence Theorem, both bounded monotonic subsequences converge: $\lim_{n\to\infty} s_{2n} = L_1$ and $\lim_{n\to\infty} s_{2n+1} = L_2$. Finally,

$$L_1 - L_2 = \lim_{n\to\infty}(s_{2n} - s_{2n+1}) = \lim_{n\to\infty} a_{2n+1} = 0$$

So $L_1 = L_2 = L$. Since both the even and odd subsequences converge to the same limit $L$, the entire sequence of partial sums $(s_n)$ converges to $L$, thus the series converges. □

**Theorem 2.36.** ***Cauchy Criterion:*** *The series $\sum a_n$ converges iff*

$$\forall \epsilon > 0 \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N : |\sum_{k=n}^{m} a_k| < \epsilon$$

**Theorem 2.37.** ***Root Criterion*** *(Wurzelkriterium): Let $\rho = \limsup \sqrt[n]{|a_n|}$. If $\rho < 1$, abs conv; if $\rho > 1$, div.*

**Theorem 2.38.** ***Ratio Criterion*** *(Quotientenkriterium): Let $\rho = \lim \left|\frac{a_{n+1}}{a_n}\right|$. If $\rho < 1$, abs conv; if $\rho > 1$, div.*

**Example 2.39.**

$$\sum_{n=1}^{\infty} n^4 e^{-n}$$

$$\rho = \lim_{n\to\infty} \frac{(n+1)^4 e^{-(n+1)}}{n^4 e^{-n}} = \lim_{n\to\infty} \left(\frac{n+1}{n}\right)^4 e^{-1} = e^{-1} < 1$$

*Therfore the series converges.*

**Definition 2.40.** *A **power series** (Potenzreihe) centered at $a$ has the form*

$$\sum_{k=0}^{\infty} c_k (x-a)^k$$

**Theorem 2.41.** *Every power series has a **convergence radius** $R$, with the **interval of convergence** $(a-R, a+R)$. $R$ is computed by the formula $R = \frac{1}{\limsup |c_n|^{1/n}}$.*

**Example 2.42.**

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} x^n$$

$$\rho = \limsup \frac{1}{n^{1/n}} = 1 \implies R = 1$$

*Therfore the series converges for*

$$x \in (-1, 1)$$

.

**Remark 2.43.** *For $x = 1$ we have the harmonic series, which diverges. For $x = -1$ we have the alternating harmonic series, which converges.*

# 3 Functions

**Definition 3.1.** *A **function** $f : D \to C$ is a map from a domian $D$ to a codomain $C$ s.t. for each $x \in D$ there is exactly one $y \in C$ st. $f(x) = y$.*

**Remark 3.2.** *The input is called the **independent variable** (argument) while the output is called the **dependent variable**.*

**Definition 3.3.** *A function $f$ is called **bouned** if there exists $M \in \mathbb{R}$ such that $|f(x)| \leq (\geq) M$ for all $x \in D$.*

**Definition 3.4.** *A function $f$ is called*

- ***injective** iff $\forall x_1, x_2 \in A : f(x_1) = f(x_2) \implies x_1 = x_2$.*
- ***surjective** iff $\forall y \in B \exists x \in A : f(x) = y$.*
- ***bijective** iff $f$ is injective and surjective.*

**Example 3.5.** *Let $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^2$. Observe that $f$ is not surjective as no output is negative, nor is it injective as for ex. $f(1) = f(-1) = 1$. But $g : \mathbb{R}_0^+ \to \mathbb{R}_0^+$ with $g(x) = x^2$ is both injective and surjective.*

**Definition 3.6.** *The **restriction** of $f$ under $D' \subset \mathbb{D}$ is defined as $f \upharpoonright_{D'} : D' \to C, x \mapsto f(x) \quad \forall x \in \mathbb{D}$.*

**Remark 3.7.** *Note that $f \upharpoonright_{D'}$ and $f$ are two diffrent functions.*

**Definition 3.8.** *Let $f : D \to C$ and $g : C \to B$ be functions then*

$$g \circ f : D \to B \quad \text{with} \quad g \circ f(x) = g(f(x)) \quad \forall x \in D$$

*is called the **composition** of $f$ and $g$.*

**Definition 3.9.** *The **identity function** $id_D : D \to D$ is defined as $id_D(x) = x \quad \forall x \in D$.*

**Definition 3.10.** *If a function $f : D \to C$ is bijective then there exists a unique function $f^{-1} : C \to D$ such that*

$$f(f^{-1}(y)) = id_C = y \quad \forall y \in C$$

*and*

$$f^{-1}(f(x)) = id_D = x \quad \forall x \in D$$

*$f^{-1}$ is called the* ***inverse function*** *of $f$.*

**Definition 3.11.** *A function $f : \mathbb{R} \to \mathbb{R}$ is called*

- ***even*** *iff $\forall x \in \mathbb{R} : f(-x) = f(x)$*
- ***odd*** *iff $\forall x \in \mathbb{R} : f(-x) = -f(x)$*

**Example 3.12.** *The function $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = x^3$ is odd as $f(-x) = (-x)^3 = -x^3 = -f(x)$ while the function $g : \mathbb{R} \to \mathbb{R}$ with $g(x) = x^2$ is even as $g(-x) = (-x)^2 = x^2 = g(x)$.*

**Definition 3.13.** *A function $f : \mathbb{R} \to \mathbb{R}$ is called*

- ***(strictly) monotonically increasing*** *iff $\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) < f(x_2)$*
- ***(strictly) monotonically decreasing*** *iff $\forall x_1, x_2 \in \mathbb{R} : x_1 < x_2 \implies f(x_1) > f(x_2)$*
- ***(strictly) monotonic*** *iff $f$ is either (strictly) monotonically increasing or (strictly) monotonically decreasing*

**Theorem 3.14.** *Every strictly monotonic function is injective.*

*Proof.* Let $f$ be strictly monotonic (wlog increasing) and assume it is not injective then there exists $x_1 \neq x_2$ such tat $f(x_1) = f(x_2)$. Wlog assume $x_1 < x_2$ then $f(x_1) < f(x_2)$ which contradicts the assumption that $f(x_1) = f(x_2)$. □

**Remark 3.15.** *Observe that functions do not have to be even or odd.*

**Definition 3.16.** *The* ***graph*** *of a function $f : D \to C$ is the set of all ordered pairs $(x, f(x))$ for $x \in D$. It is* ***konvex*** *if it lies below the line segment connecting any two points on the graph. Similarly it is called* ***konkav*** *if it lies above the line segment connecting any two points on the graph.*

## 3.1 Limits in the context of functions

**Definition 3.17.** *Let $f : \mathbb{D} \to \mathbb{R}$ be a function satisfying $\mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$ for all $\delta > 0$ and $x_0 \in \mathbb{R}$. We say that $\lim_{x \to x_0} f(x) = L$ iff*

$$\forall \varepsilon > 0 \exists \delta > 0 s.t.$$

$$x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta) \implies |f(x) - L| < \varepsilon$$

*The limit $L$ is unique.*

**Remark 3.18.** *The intuition behind the definition of the limit is that for any given distance $\varepsilon > 0$ around $L$, we can find a $\delta > 0$ such that for all $x \in \mathbb{D}$ that are at least $\delta$ close to $x_0$, we have $|f(x) - L| < \varepsilon$. So for any boound on the output ($\varepsilon$) there exists a bound on the input ($\delta$) such that all elements in the input bound $(x_0 - \delta, x_0 + \delta)$ are mapped to the output bound $(L - \varepsilon, L + \varepsilon)$.*

**Definition 3.19.** *A function $f : \mathbb{D} \to \mathbb{R}$ has in $x_0$ a*

- ***right-sided limit*** $\lim_{x \searrow x_0} f(x) = \lim_{x \to x_0^+} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists \delta > 0 \quad s.t. \quad \forall x \in \mathbb{D}(x_0 < x < x_0 + \delta \implies |f(x) - L| < \varepsilon)$$

- ***left-sided limit*** $\lim_{x \nearrow x_0} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists \delta > 0 \quad s.t. \quad \forall x \in \mathbb{D}(x_0 - \delta < x < x_0 \implies |f(x) - L| < \varepsilon)$$

**Definition 3.20.** *A function $f : \mathbb{D} \to \mathbb{R}$ has a*

- ***limit for x approaching infinity*** $\lim_{x \to \infty} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists M > 0 \quad s.t. \quad \forall x \in \mathbb{D}(x > M \implies |f(x) - L| < \varepsilon)$$

- ***limit for x approaching negative infinity*** $\lim_{x \to -\infty} f(x) = L$ *iff*

$$\forall \varepsilon > 0 \exists M > 0 \quad s.t. \quad \forall x \in \mathbb{D}(x < M \implies |f(x) - L| < \varepsilon)$$

**Definition 3.21.** *A function $f : \mathbb{D} \to \mathbb{R}$ has in $x_0$ a*

- ***improper limit approaching infinity*** $\lim_{x \to x_0} f(x) = \infty$ *iff*

$$\forall N > 0 \exists \delta > 0 \quad s.t. \quad \forall x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) > N)$$

- ***improper limit approaching negative infinity*** $\lim_{x\to x_0} f(x) = -\infty$ *iff*

$$\forall N > 0 \exists \delta > 0 \quad s.t. \quad \forall x \in \mathbb{D}(0 < |x - x_0| < \delta \implies f(x) < -N)$$

**Theorem 3.22.** *Let $A \in \mathbb{R}$ and assume $\lim_{x\to x_0} f(x) = K(\neq \pm\infty)$ and $\lim_{x\to x_0} g(x) = L(\neq \pm\infty)$.*

- $\lim_{x\to x_0}(f(x) + g(x)) = K + L$
- $\lim_{x\to x_0}(f(x) - g(x)) = K - L$
- $\lim_{x\to x_0}(f(x)g(x)) = KL$
- $\lim_{x\to x_0}(f(x)/g(x)) = K/L$
- $\lim_{x\to x_0}(Af(x)) = AK$
- $\lim_{x\to x_0}(f(x)/g(x)) = K/L$ *if* $L \neq 0$

## 3.2 Continuity of functions

**Definition 3.23.** *A function $f : D \to \mathbb{R}$ is called **continous** at $x_0$ if*

$$\forall \epsilon > 0 \exists \delta > 0 \text{ such that } \forall x \in I(\mid x - x_0 \mid < \delta \implies \mid f(x) - f(c) \mid > \epsilon)$$

*The function is called continuous if it is continuous at every point in its domain.*

**Remark 3.24.** *The **intuition** behind this definition is that for any given distance $\epsilon > 0$ around $f(x_0)$, we can find a $\delta > 0$ such that for all $x \in D$ that are at least $\delta$ close to $x_0$, we have $|f(x) - f(x_0)| < \epsilon$. So for any bound on the output ($\epsilon$) there exists a bound on the input ($\delta$) such that all elements in the input bound $(x_0 - \delta, x_0 + \delta)$ are mapped to the output bound $(f(x_0) - \epsilon, f(x_0) + \epsilon)$.*

**Definition 3.25.** *Let $f : D \to \mathbb{R}$ be a function. Then $f$ is called **left(right)-continuous** at $x_0$ if*

$$\lim_{x\to x_0^{-(+)}} f(x) = f(x_0)$$

**Theorem 3.26.** *Let $f : D \subset \mathbb{D}(f) \subset \mathbb{R}$ be a function and $x_0 \in D$. $f$ is continuous at $x_0$ iff for every sequence $(x_n)_{n\in\mathbb{N}} \subset D$ with $\lim_{n\to\infty} x_n = x_0$ it holds that $\lim_{n\to\infty} f(x_n) = f(x_0)$.*

**Remark 3.27.** *Continuity can also be viewed in terms of intervals, where a function $f$ is continuous iff for every open interval $I \subset \mathbb{R}$ it holds that $f^{-1}(I)$ is open.*

**Lemma 3.28.** *Let $f, g$ be continuous functions, the follwing holds:*

- *$f + g$ is continuous*
- *$f - g$ is continuous*
- *$f \cdot g$ is continuous*
- *$c \cdot f$ is continuous for all $c \in \mathbb{R}$*
- *$\frac{f}{g}$ is continuous if $g(x) \neq 0$*
- *$f \circ g$ is continuous*

*Proof.* Addition/Subtraction/Multiplication/Division follow from the sequence-definition of continuity and the properties of limits. It is left to prove that $f \circ g$ is continuous if $f$ and $g$ are. Let $g$ be continuous at a point $c$, and let $f$ be continuous at the point $g(c)$.

By the limit definition of continuity, since $g$ is continuous at $c$, we know:

$$\lim_{x \to c} g(x) = g(c)$$

Similarly, since $f$ is continuous at $g(c)$, for any variable $y$ approaching $g(c)$, we know:

$$\lim_{y \to g(c)} f(y) = f(g(c))$$

To find the limit of the composite function $f(g(x))$ as $x \to c$, we substitute $g(x)$ for our input $y$. Because $f$ is continuous, we are allowed to pass the limit directly inside the function $f$:

$$\lim_{x \to c} f(g(x)) = f\left(\lim_{x \to c} g(x)\right)$$

Now, we just substitute the known limit of $g(x)$:

$$f\left(\lim_{x \to c} g(x)\right) = f(g(c))$$

Therefore, $\lim_{x \to c}(f \circ g)(x) = (f \circ g)(c)$. This perfectly satisfies the definition of continuity, proving that $f \circ g$ is continuous at $c$ □

**Theorem 3.29.** *Let $I$ be an interval and let $f : I \to J$ be a continuous and bijective function. Then $J$ is an interval too and the inverse function $f^{-1} : J \to I$ is continuous.*

**Theorem 3.30.** *Let $f : D \to \mathbb{R}$ be a continuous function. Let $c \in [f(a), f(b)]$ then there exists a $x_0 \in [a, b]$ such that $f(x_0) = c$.*

*Proof.* Let $f : [a, b] \to \mathbb{R}$ be a continuous function. Without loss of generality, assume $f(a) < f(b)$. We choose an arbitrary $c \in \mathbb{R}$ such that $f(a) \leq c \leq f(b)$ and define the set:

$$X := \{t \in [a, b] \mid f(t) \leq c\}$$

Observe that $X \neq \emptyset$ (since $a \in X$ as $f(a) \leq c$) and $X$ is bounded above by $b$. Thus, by the completeness axiom of the real numbers, the supremum $s := \sup X$ exists, and clearly $s \in [a, b]$.

By the definition of the supremum, for all $n \in \mathbb{N}$, there exists $t_n \in X \cap [s - \frac{1}{n}, s]$. Because $t_n \to s$ as $n \to \infty$ and $f$ is continuous, we have $\lim_{n\to\infty} f(t_n) = f(s)$. Since $t_n \in X$, we know $f(t_n) \leq c$ for all $n \in \mathbb{N}$. By the limit-preserving properties of inequalities, it follows that $f(s) \leq c$.

Now, assume for the sake of contradiction that $f(s) < c$. Since $f(s) < c \leq f(b)$, we know $s \neq b$, which implies $s < b$.

Let $\epsilon = c - f(s) > 0$. By the continuity of $f$ at $s$, there exists a $\delta > 0$ such that for all $y \in [a, b]$:

$$|y - s| < \delta \implies |f(y) - f(s)| < \epsilon$$

This implies $f(y) < f(s) + \epsilon = c$.

Now, define $y := \min(b, s+\frac{\delta}{2})$. Since $s < b$ and $\delta > 0$, we strictly have $y > s$. Furthermore, $y \in [a, b]$ and $|y - s| = \frac{\delta}{2} < \delta$. Therefore, by our continuity implication, $f(y) < c$, which means $y \in X$.

However, we have found an element $y \in X$ such that $y > s$, which directly contradicts the fact that $s$ is an upper bound (the supremum) of $X$.

Thus, our assumption that $f(s) < c$ must be false. Since we have already established that $f(s) \leq c$, we conclude that $f(s) = c$. $\square$

**Definition 3.31.** *A bounded and closed interval is called **compact**.*

**Example 3.32.** *The interval $[0, 1]$ is compact while $[0, \infty)$ is not.*

**Lemma 3.33.** *Let $[a, b]$ be a compact interval and let $(x_n)_{n\in\mathbb{N}} \subset [a, b]$ be a sequence. Then there exists a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ such that*

$$\lim_{k\to\infty} x_{n_k} = x_0 \text{ for some } x_0 \in [a, b].$$

**Definition 3.34.** *Let $f : D \subset \mathbb{R} \to \mathbb{R}$ be a function*

- *If for $x_0 \in D$ it holds that $f(x_0) \geq f(x) \forall x \in D$, then $f$ has a local maximum at $x_0$.*

- *If for* $x_0 \in D$ *it holds that* $f(x_0) \leq f(x) \forall x \in D$, *then* $f$ *has a local minimum at* $x_0$.
- *Local maxima and minima are called local extrema.*

**Theorem 3.35.** *Let* $f : I \subset \mathbb{R} \to \mathbb{R}$ *be a continous function and* $I$ *be a compact interval. Then* $f$ *is bounded and attains its maximum and minimum value on* $I$. *Hence*

$$\exists x_{min}, x_{max} \in I \text{ such that } f(x_{min}) \leq f(x) \leq f(x_{max}) \forall x \in I$$

# 4 Differential calculus

**Definition 4.1.** *The* ***average rate of change*** *of a function $f$ in the domain between $a$ and $a+h$ is defined as*

$$\frac{\Delta y}{\Delta x} = \frac{f(a+h) - f(a)}{h}$$

*and also called the* ***difference quotient****.*

**Definition 4.2.** *The* ***derivative*** *$f'$ of a function $f$ at the coordinate $a$ is defined as*

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h} = lim_{b \to a} \frac{f(b) - f(a)}{b - a} = \frac{df}{dx}|_{x=a} = f'(a)$$

**Remark 4.3.** *The derivative $f'(a)$ can geometrically interpreted as the incline/decline of the graph of $f$ at the point $(a, f(a))$.*

**Definition 4.4.** *If for a function $f$ at the cooridnate $a$ the derivative $f'(a)$ exists then $f$ is called* ***differentiable*** *at $a$. If $f$ is differentiable for every $a \in \mathbb{D}$ then $f$ is called differentiable.*

**Example 4.5.** *let $f(x) = |x|$ we can see that for $a = 0$ there is no derivative (This can be seen in the graph as a "sharp" corner)*

**Remark 4.6.** *We can generally expect that there are no derivatives at undefined points $a \notin \mathbb{D}$.*

**Lemma 4.7.** *If $f$ is differentiable at $x_0$ then $f$ is continous at $x_0$.*

*Proof.* Assume $f$ differentiable then

$$\lim_{x \to x_0} f(x) = \lim_{h \to 0} f(x_0 + h) = \lim_{h \to 0} f(x_0 + h) - f(x_0) + f(x_0)$$

$$= f(x_0) + \lim_{h \to 0} f(x_0 + h) - f(x_0) = f(x_0) + \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} \cdot h$$

$$= f(x_0) + \lim_{h \to 0} h \cdot \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

Observe that the last term exists by the assumption hence

$$= f(x_0) + 0 \cdot f'(x_0) = f(x_0)$$

□

**Definition 4.8.** *If the limit*

$$\lim_{h>0\to 0} \frac{f(a+h)-f(a)}{h}$$

*existst then we call it the* ***right-sided derivative*** *(or the left-sided derivative analogue)*

**Definition 4.9.** *For a function $f$ we can iteratively define higher derivatives for $n \in \mathbb{N}_0$*

$$f^(0) = f, f^(1) = f', f^(2) = f'', \ldots f^(n+1) = (f^(n))'$$

*if $f^(n)$ exists we call it $n$**-times differentiable**.*

**Definition 4.10.** *if $f^(n)$ is continous we call $f$ $n$**-times continous differentiable**. The set of all n-times continous differentiable functions on $D$ is denoted as $C^n(D)$*

**Definition 4.11.** *We call $C^\infty(D) := \cap_{n=0}^{\infty} C^n(D)$ the set of* ***smooth functions****.*

**Definition 4.12.** *If for a function $f$ and some $x_0 \in \mathbb{D}$ and $\delta > 0$ we have*

$$f(x_0) \geq (\leq) f(x) \quad \forall x \in \mathbb{D} \cap (x_0 - \delta, x_0 + \delta)$$

*then $x_0$ is called a* ***local maximum (minimum)****.*

## 4.1 Derivation rules

**Theorem 4.13.** *For functions $f(x)$ we have*

- $f(x) = ax^p \implies f'(x) = apx^{p-1}$
- $f(x) = a^x \implies f'(x) = ln(a) \cdot a^x$
- $f(x) = sin(x) \implies f'(x) = cos(x)$
- $f(x) = cos(x) \implies f'(x) = -sin(x)$
- $f(x) = tan(x) \implies f'(x) = \frac{1}{cos^2(x)} = 1 + tan^2(x)$
- $f(x) = sinh(x) \implies f'(x) = cosh(x)$
- $f(x) = cosh(x) \implies f'(x) = sinh(x)$
- $f(x) = ln(x) \implies f'(x) = \frac{1}{x}$

- $f(x) = arcsin(x) \implies f'(x) = \frac{1}{\sqrt{1-x^2}}$
- $f(x) = arccos(x) \implies f'(x) = \frac{-1}{\sqrt{1-x^2}}$
- $f(x) = arctan(x) \implies f'(x) = \frac{1}{1+x^2}$

**Theorem 4.14.** ***Sum Rule:*** *For the n-th derivative of a sum:*

$$(f+g)^{(n)} = f^{(n)} + g^{(n)}$$

*Proof.* This can be proved inductively by applying the definition of the derivative $n$ times.

$$(f+g)'(x) = lim_{h\to 0}\frac{f(x+h)+g(x+h)-f(x)-g(x)}{h}$$
$$= lim_{h\to 0}\frac{f(x+h)-f(x)}{h} + lim_{h\to 0}\frac{g(x+h)-g(x)}{h}$$
$$= f'(x) + g'(x)$$

For $n = 2$ we have

$$(f+g)''(x) = (f'+g')'(x) = f''(x) + g''(x)$$

and so on. □

**Theorem 4.15.** ***General Leibniz Rule (Product):*** *The n-th derivative of a product is given by:*

$$(f\cdot g)^{(n)} = \sum_{k=0}^{n}\binom{n}{k} f^{(n-k)}g^{(k)}$$

*Proof.* This is proved analogously to the sum rule.

$$(f\cdot g)'(x) = lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x)}{h}$$
$$= lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x+h)+f(x)g(x+h)-f(x)g(x)}{h}$$
$$= lim_{h\to 0}\frac{g(x+h)(f(x+h)-f(x))}{h} + lim_{h\to 0}\frac{f(x)(g(x+h)-g(x))}{h}$$
$$= g(x)f'(x) + f(x)g'(x)$$

For $n = 2$ we have

$$(f\cdot g)''(x) = (f'g+fg')'(x) = f''g + f'g' + f'g' + fg'' = f''g + 2f'g' + fg''$$

and so on. □

**Theorem 4.16.** ***Chain Rule:*** *For the composition of functions:*

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

*Proof.* Observe that

$$g(y) = g(y_0) + g'(y_0)(y - y_0) + g(y) - g(y_0) - g'(y_0)(y - y_0)$$

with the first two summands being a linear (tangential) approximation of $g(y)$ near $y_0$

$$g(y) \approx g(y_0) + g'(y_0)(y - y_0)$$

By taking out $(y - y_0)$ of the last two summands in the original equation we get:

$$g(y) = g(y_0) + g'(y_0)(y - y_0) + \left[\frac{g(y) - g(y_0)}{y - y_0} - g'(y_0)\right](y - y_0)$$

We define $\epsilon = \frac{g(y)-g(y_0)}{y-y_0} - g'(y_0)$ and get

$$\lim_{x \to x_0} \frac{g(f(x)) - g(f(x_0))}{x - x_0}$$

$$= \lim_{x \to x_0} \left(f(f(x_0)) + g'(f(x_0))[f(x) - f(x_0)] + \epsilon f(x)[f(x) - f(x_0)] - g(f(x_0))\right) \frac{1}{x - x_0}$$

$$= g'(y_0) f'(x_0)$$

□

**Theorem 4.17.** ***Quotient Rule:*** *For* $g(x) \neq 0$*:*

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

**Remark 4.18.** *This follows directly from the product rule.*

**Theorem 4.19.** ***Inverse Functions:*** *If* $f$ *is differentiable and injective:*

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

## 4.2 Bernoulli-L'Hopital

**Theorem 4.20.** *Let* $I \subset \mathbb{R}$ *be a an intervall,* $f : I \to \mathbb{R}$*, and let* $x_0$ *be a local Extrema of* $f$*. Then one of the following holds:*

- $x_0$ *is an endpoint of* $I$
- $f$ *is not differentiable at* $x_0$

- *$f$ is differentiable at $x_0$ and $f'(x_0) = 0$*

**Remark 4.21.** *From $f'(x_0) = 0$ we can not conclude that $x_0$ is an extremum.*

**Theorem 4.22.** ***Rolle's Theorem:*** *Let $a < b$ with $f(a) = f(b)$ and $f : [a, b] \to \mathbb{R}$ a completely continous function on $[a, b]$ and differentiable on $(a, b)$ then $\exists \xi \in (a, b)$ with $f'(\xi) = 0$*

*Proof.* As $[a, b]$ is compact we can conclud that there exists $\xi$ such that $f(\xi)$ is max or min hence $f'(\xi) = 0$. If $\xi \in (a, b)$ we found our $\xi$ else the max and min is at $a$ therfore also at $b$ hence the function is constant which implies that $f'(x) = 0 \forall x \in [a, b]$. □

**Theorem 4.23.** ***Mean-Value-Theorem:*** *Let $a < b, f : [a, b] \to \mathbb{R}$ a completely continous function on $[a, b]$ and differentiable on $(a, b)$ then $\exists \xi \in (a, b)$ with*

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}$$

*Proof.* We define a function $g$ with

$$g(x) \coloneqq f(x) - \frac{f(b) - f(a)}{b - a}(x - a)$$

and observe that $g$ fulfilles the prequisites to apply rolles theorem. Hence

$$\begin{aligned} \exists \xi g'(\xi) = 0 \\ \implies \exists \xi f'(\xi) - \frac{f(b) - f(a)}{b - a}\xi = 0 \\ \implies \exists \xi f'(\xi) = \frac{f(b) - f(a)}{b - a} \end{aligned}$$

□

**Theorem 4.24.** ***Cauchy's Mean-Value-Theorem:*** *Let $a < b, f : [a, b] \to \mathbb{R}$ a completely continous function on $[a, b]$ and differentiable on $(a, b)$ then $\exists \xi \in (a, b)$ with*

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}$$

*Proof.* We define a function

$$F(x) \coloneqq g(x)(f(b) - f(a)) - f(x)(g(b) - g(a))$$

and observe that $F$ fulfilles the prequisites to apply rolles theorem. Hence

$$\begin{aligned} &\exists \xi F'(\xi) = 0 \\ \implies &\exists \xi g'(\xi)(f(b) - f(a)) - f'(\xi)(g(b) - g(a)) = 0 \\ \implies &\exists \xi \frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)} \end{aligned}$$

□

**Theorem 4.25.** ***Bernoulli-L'Hopital's Rule:*** *Let $f, g : I \to \mathbb{R}$ be differentiable functions on $(a, b)$ and let $x_0 \in (a, b)$. If $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in (a, b)$. If*

$$\lim_{x \to x_0} |f(x)| = \lim_{x \to x_0} |g(x)| = 0 or \infty \quad and \quad \exists L = \lim_{x \to x_0} \frac{f'(x)}{g'(x)}$$

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)} = L$$

**Example 4.26.** *Let $f(x) = sin(x)$ and $g(x) = x$ we have $sin(0) = 0$ and $g(0) = 0$ further we have $f'(x) = cos(x)$ and $g'(x) = 1 \neq 0$ hence*

$$\lim_{x \to 0} \frac{sin(x)}{x} = \lim_{x \to 0} \frac{cos(x)}{1} = 1$$

**Remark 4.27.** *This example can also be solved via the definition of the derivative.*

$$\lim_{x \to 0} \frac{sin(x)}{x} = \lim_{x \to 0} \frac{sin(x) - sin(0)}{x - 0} = sin'(0) = cos(0) = 1$$

## 4.3 Convex and Concave graphs

**Definition 4.28.** *A function $f : I \to \mathbb{R}$ is called **convex** iff*

$$\forall a, b \in I \forall t \in (0, 1) : f((1 - t)a + bt) \leq (1 - t)f(a) + tf(b)$$

**Definition 4.29.** *A function $f : I \to \mathbb{R}$ is called **concave** iff*

$$\forall a, b \in I \forall t \in (0, 1) : f((1 - t)a + bt) \geq (1 - t)f(a) + tf(b)$$

**Theorem 4.30.** *Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \to \mathbb{R}$ be differentiable. Then*

$$\forall x : f'(x) \geq 0 \iff f \text{ is increasing}$$

**Remark 4.31.** *If $f'(x) = 0$ for all $x$ then $f$ is constant.*

**Theorem 4.32.** *Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \to \mathbb{R}$ be differentiable. Then*

$$f \text{ is convex} \quad \Longleftrightarrow \quad f' \text{ is increasing}$$

**Theorem 4.33.** *Let $(a, b) \subset \mathbb{R}$ and $f : (a, b) \to \mathbb{R}$ be twice differentiable. Then*

$$f \text{ is convex} \quad \Longleftrightarrow \quad f'' \geq 0$$

## 4.4 Approximation of functions

**Lemma 4.34.** *We can localy approximate the graph of $f$ at $x_0$ with the tangential line $t(x)$*

$$f(x) \approx t(x) = f(x_0) + f'(x_0)(x - x_0)$$

**Example 4.35.** *Let $f(x) = x^2$ and $x_0 = 2$ then*

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0) = 2^2 + 2(2)(x - 2) = 4 + 4(x - 2) = 4x - 4$$

**Lemma 4.36.** *To even more accurately approximate the function $f$ we can add further terms to the tangential line expansion*

$$f(x) \approx t(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2$$

**Remark 4.37.** *The* ***Intuition*** *behind this is that we want a function that looks something like this:*

$$P(x) = A + B(x - x_0) + C(x - x_0)^2$$

*if $P(x_0) = f(x_0)$ we have $A = f(x_0)$. Further if $P'(x_0) = f'(x_0)$ we have $B = f'(x_0)$ and if $P''(x_0) = f''(x_0)$ we have $C = f''(x_0)/2$.*

**Example 4.38.** *Let $f(x) = x^2$ and $x_0 = 2$ then*

$$f(x) \approx f(x_0)+f'(x_0)(x-x_0)+\frac{f''(x_0)}{2}(x-x_0)^2 = 2^2+2(2)(x-2)+\frac{2}{2}(x-2)^2 = 4+4(x-2)+(x-2)^2$$

**Theorem 4.39.** ***Taylor's Theorem:*** *Let $f : (a,b) \to \mathbb{R}$ be $n+1$ times differentiable. Then*

$$f(x) = P_n(x) + R(x)$$

*with $P_n(x)$ the Taylor Polynomial of degree $n$ and $R(x)$ the remainder term*

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

$$R(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1} \quad \text{for some } \xi \in (x_0, x) \text{ or } (x, x_0)$$

**Remark 4.40.** *There are other forms of the remainder term, such as the Lagrange form and the Cauchy form.*

**Example 4.41.** *Assume we know that the equation*

$$\sqrt{3x-4} + 2\sin x - x = \frac{1}{2}$$

*has a solution near $x_0 = 2$. We can use Taylor's Theorem to approximate the solution. Let $f(x) = \sqrt{3x-4} + 2\sin x - x$. Then*

$$f(2) = \sqrt{2} + 2\sin 2 - 2$$

$$f'(2) = \frac{3}{2\sqrt{2}} + 2\cos 2 - 1$$

*Therefore the Taylor Approximation up to degree 2 is*

$$f(x) \approx f(2) + f'(2)(x-2)$$

*Setting this to $\frac{1}{2}$ and solving for $x$ gives us the linear approximation of the solution.*

**Definition 4.42.** *The* ***Taylor series expansion*** *of an infinitely differentiable function $f : (a,b) \to \mathbb{R}$ at $x_0$ is given by*

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

**Example 4.43.** *The following Taylor series converge for all $x \in \mathbb{R}$:*

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots$$

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$$

$$\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$$

**Important 4.44.** *Not all infinitely differentiable functions have a Taylor series expansion that converges to the function itself. For example $ln(1+x)$, $\frac{1}{1-x}$, $(1+x)^p$ only converge for $-1 < x < 1$*

**Theorem 4.45.** *Let $f(x)$ and $g(x)$ be two powerseries with $x$ in their Radius of Convergence $R$.*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$g(x) = \sum_{n=0}^{\infty} b_n x^n$$

*Then*

$$f(x)g(x) = \sum_{n=0}^{\infty} \left( \sum_{k=0}^{n} a_k b_{n-k} \right) x^n$$

**Theorem 4.46.** *Let $f(x)$ be a power series with $x$ in its Radius of Convergence $R$.*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

*then*

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

**Theorem 4.47.** *Let $f(x)$ be a power series with $x$ in its Radius of Convergence $R$.*

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

*then*

$$\int f(x) dx = \sum_{n=0}^{\infty} \int a_n x^n dx$$

**Example 4.48.** *Assume we know $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ for $-1 < x < 1$ and we want to calculate the Taylor-Expansion of $g(x) = (1-x)^{-2}$. We observe that $g(x) = (\frac{d}{dx})(1-x)^{-1}$. Therefore*

$$g(x) = \frac{d}{dx} \sum_{n=0}^{\infty} x^n = \sum_{n=1}^{\infty} n x^{n-1} = \sum_{n=0}^{\infty} (n+1) x^n$$

# 5 Differential equations

## 5.1 Modelling of differential equations

**Example 5.1.** *Assume at time $t = 0$ one takes in a certain amount of alcohol. The alcohol first flows to the stomac and is then absorbed into the blood. We want to find a description of the current amount of alcohol in the stomac $m_S(t)$ and in the blood $m_B(t)$.*

*We assume the following:*

- *the amount of alcohol in the stomac decreases by $\alpha \Delta t m_S(t)$*

$$m_S(t + \Delta t) = m_S(t) - \alpha \Delta t \cdot m_S(t)$$
$$\implies \frac{m_S(t + \Delta t) - m_S(t)}{\Delta t} = -\alpha \cdot m_S(t)$$
$$\implies m_S'(t) = -\alpha \cdot m_S(t)$$

- *the amount of alcohol in the blood is decreased by a constant amount of $\beta \Delta t$*

$$m_B(t)' = \alpha \cdot m_S(t) - \beta$$

**Definition 5.2.** *A differential equation is an equation which contains derivatives of functions.*

**Definition 5.3.** *A differential equation is called*

- *of **n-th order**, if the n-th derivative is the highest occurring derivative.*
- ***linear**, if the unknown function $u(t)$ and its derivatives appear in the equation in the form of a linear combination.*
- ***homogeneous**, if it only contains multiples of the unknown function $u(t)$ and its derivatives.*

**Definition 5.4.** *An ordinary differential equation (ODE) is a differential equation which contains only derivatives of one independent variable.*

## 5.2 Linear, homogeneous ODEs

**Theorem 5.5.** ***Superposition principle**: If $y_1$ and $y_2$ are solutions to a linear homogeneous differential equation so is*

$$y(t) = c_1 y_1(t) + c_2 y_2(t), \quad c_1, c_2 \in \mathbb{R}$$

**Theorem 5.6.** *Linear homogenous differential equations of $n$-th order have $n$ solutions $y_1, ..., y_n$ such that*

$$y(x) = C_1 y_1(x) + ... + C_n y_n(x), \quad C_i \in \mathbb{R}, \text{ for } i = 1, ..., n$$

*is called the general solution.*

**Lemma 5.7.** *Let $y(x) = e^{\lambda x}, \lambda \in \mathbb{R}$ then by deriving $n$ times and taking out $e^{\lambda x}$ we will get the characteristic equation*

$$a_n \lambda^n + a_{n-1} \lambda^{n-1} + ... + a_1 \lambda + a_0 = 0$$

*And the characteristic polynom*

$$P(\lambda) = a_n \lambda^n + ... + a_0$$

**Theorem 5.8.** *We can find the general solution of a linear, homogeneous ODE of $n$-th order by finding the roots of the characteristic polynom*

- ***Distinct real roots**: $y_i(x) = e^{\lambda_i x}, \lambda_i \in \mathbb{R}$*
- ***Multiple real roots**: $y_i(x) = x^k e^{\lambda_i x}, \lambda_i \in \mathbb{R}$*
- ***Complex roots**: $y_i(x) = e^{\alpha x} \cos(\beta x), e^{\alpha x} \sin(\beta x), \lambda_i = \alpha + i\beta \in \mathbb{C}$*

## 5.3 Boundary-value Problems

**Remark 5.9.** *Assume we have the general solution $y = C_1 y_1 + C_2 y_2$ for a linear homogeneous ODE of second order. If we have some conditions for the function at certain points, e.g. $y(a) = A, y(b) = B$, then we can find the constants $C_1, C_2$ by solving the following system of linear equations:*

$$y(a) = C_1 y_1(a) + C_2 y_2(a)$$

$$y(b) = C_1 y_1(b) + C_2 y_2(b)$$

# 6 Integration

**Definition 6.1.** *Given a function $f : I \to \mathbb{R}$ for $I \subseteq \mathbb{R}$ an open interval. An **anti-derivative** (or **primitive function**) of $f$ is a function $F : I \to \mathbb{R}$ such that $F'(x) = f(x)$ for all $x \in I$.*

**Definition 6.2.** *The family of all anti-derivatives of a function $f : I \to \mathbb{R}$ for $I \subseteq \mathbb{R}$ an open interval is called the **indefinite integral** and is denoted by*

$$\int f(x)dx = \{F : I \to \mathbb{R} : F'(x) = f(x)\}$$

**Remark 6.3.** *For indefinite integrals we have*

$$\int f(x) \pm g(x)dx = \int f(x)dx \pm \int g(x)dx$$

$$\int cf(x)dx = c \int f(x)dx$$

$$\dots$$

*Note that to determine indefinite integrals we can use differentiation rules in reverse.*